

- 3** Marram grass, *Ammophila arenaria*, is a plant found on coastal sand dunes in Europe and North Africa.

The plant is well adapted for living on sand dunes. The leaves can grow to over one metre in length.

The photograph shows marram grass growing on sand dunes.



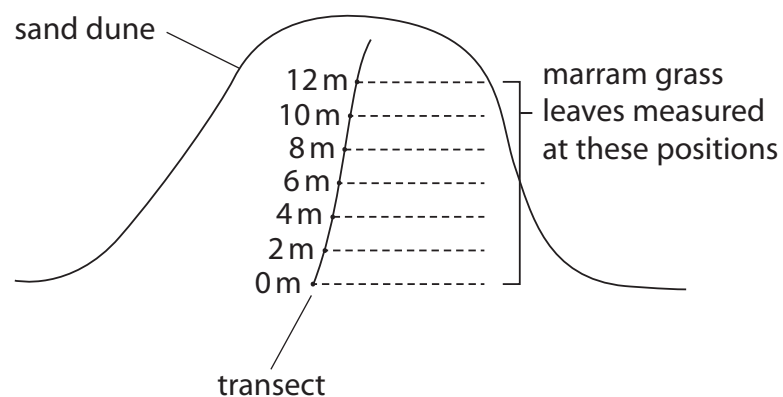
(Source: © Chris Millican)

A student observed that the leaves seemed to be longer in some parts of the dune than others.

A transect was set up from the base of the dune to the top of the dune.

At two-metre intervals along the transect, ten plants were selected. One leaf from each plant was measured using a metre rule.

The diagram shows the transect on the sand dune.



- (a) Identify one risk you might encounter when carrying out this investigation and how you could reduce this risk.

(2)

Risk

How to reduce this risk

- (b) The table shows the results of this investigation:

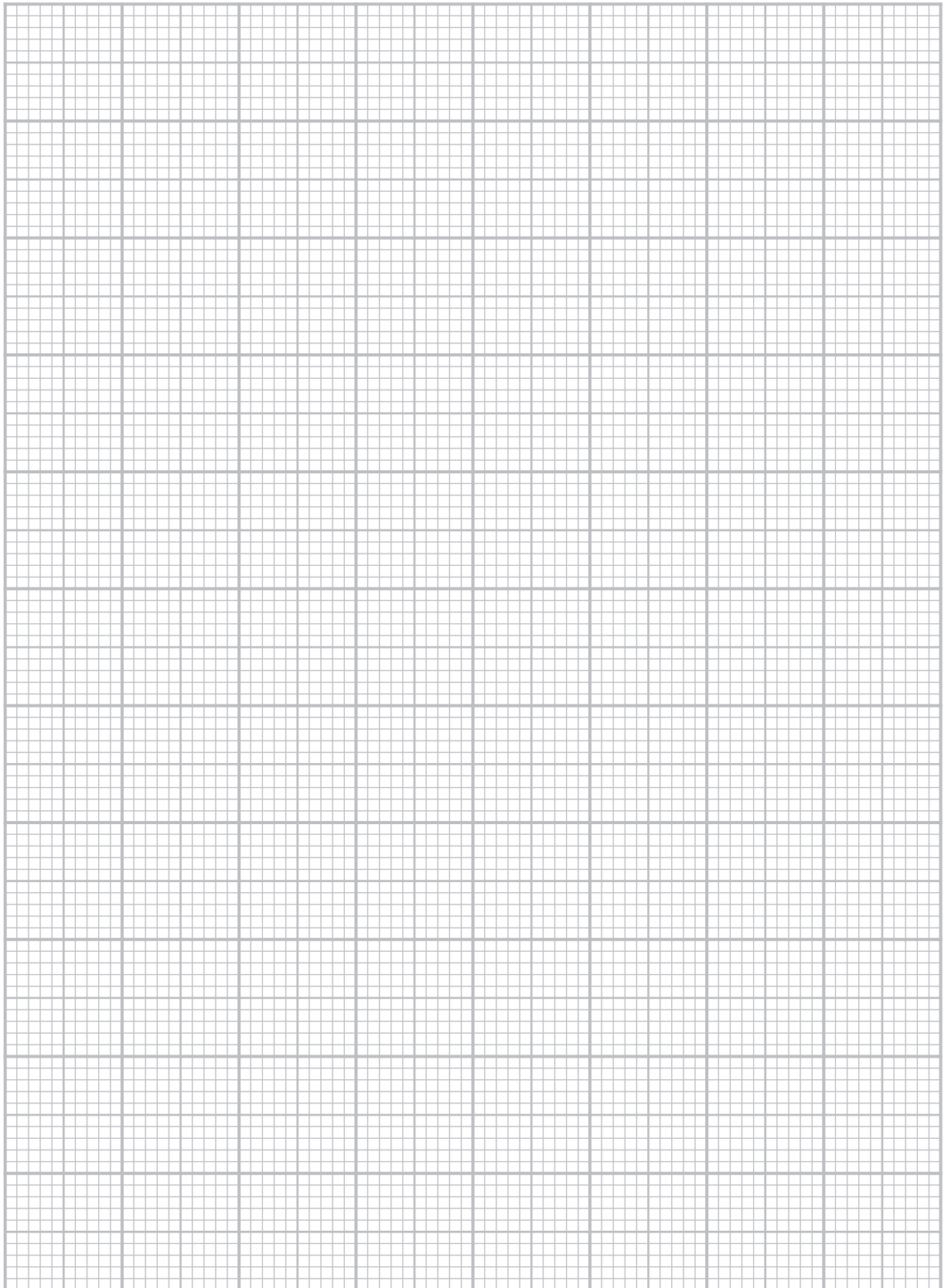
Position on dune / m	Length of leaf / cm										Mean length / cm
0	31	56	40	36	24	31	33	54	35	29	37
2	38	42	27	35	40	51	32	27	37	21	35
4	47	41	56	37	43	55	29	48	39	42	44
6	72	61	64	76	80	58	73	56	42	51	63
8	108	92	101	95	109	98	79	106	103	99	99
10	93	102	96	98	83	87	94	96	79	88	92
12	104	87	81	90	103	89	95	88	105	97	94

- (i) State a suitable null hypothesis for this investigation.

(1)

- (ii) Plot a suitable graph to show the effect of position on dune on the mean length of marram grass.

(2)



(c) The student used a statistical test to analyse the data.

To calculate the correlation coefficient, the student produced this table.

Position on dune / m (a)	Mean length of marram grass / m (b)	Rank (a)	Rank (b)	d	d^2
0	0.37	1	2	-1	
2	0.35	2	1	1	
4	0.44	3	3	0	
6	0.63	4	4	0	
8	0.99	5	7	-2	
10	0.92	6	5	1	
12	0.94	7	6	1	

(i) Calculate the correlation coefficient, r_s , using the formula:

(2)

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Where:

$\sum$ = the sum of

d = the difference between each pair of ranks

n = the size of the sample (number of pairs of values)

Answer

(ii) The table shows some critical values for this statistical test.

Number of pairs of values	Level of significance (p)		
	0.10	0.05	0.01
4	1.000	–	–
5	0.900	1.000	–
6	0.829	0.886	1.000
7	0.714	0.786	0.929
8	0.643	0.738	0.881
9	0.600	0.700	0.833
10	0.564	0.648	0.794

Explain the conclusion that can be drawn from this investigation.

Use your graph, your calculated r_s value and the table of critical values to support your answer.

(2)

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(d) Describe **three** improvements that could be made to this investigation.

(3)

(e) (i) Suggest **one** abiotic factor that might cause the difference in length of the marram grass leaves at these positions on the dune.

(1)

(ii) Describe how this factor could be measured to produce valid comparisons in this investigation. (3)

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(Total for Question 3 = 16 marks)
